

Sai Nandu Vajhala

+91 7207006151 | vajhalasainandu@gmail.com | linkedin.com/in/vajhala-sai-nandu | github.com/SaiNanduVajhala

EDUCATION

Sreyas Institute of Engineering and Technology

BTech. in Artificial Intelligence and Machine Learning

Hyderabad, Telangana

Aug. 2024 – May 2027 (*expected*)

Samskruti College of Engineering and Technology

Diploma in Electrical and Electronics Engineering

Hyderabad, Telangana

Aug. 2021 – April 2024

EXPERIENCE

Data Science Intern

Xylofy AI

June 2026 – July 2026

Remote

- Engineered preprocessing pipelines and classification models (Scikit-learn/XGBoost) for employee attrition analysis, improving prediction accuracy by 15%.
- Built a demand forecasting system comparing SARIMA, Prophet, and XGBoost (20.5% MAPE) and deployed an interactive Streamlit dashboard to optimize inventory stocking strategies.

PROJECTS

Oneiros | Cognitive Memory OS for AI Agents

July 2026

- Technologies:** Python, Cognee Cloud, SQLite, FastAPI, Three.js, React.
- Built a memory OS based on Erik Hoel's *Overfitted Brain Hypothesis*, creating an AI agent that sleeps to consolidate raw experiences and wakes up smarter with optimized, abstraction-linked knowledge schemas.
- Integrated Cognee graph database schemas with local SQLite caching for sub-millisecond dynamic entity recall and interactive 3D force-directed visualizations.

GPT-2 Code Completion Model Fine-Tuning

May 2026

- Technologies:** Python, PyTorch, Hugging Face Transformers, GPT-2, CodeXGLUE, Kaggle GPUs.
- Fine-tuned a GPT-2 transformer on the CodeXGLUE code-completion dataset (13k+ code samples) to build an autoregressive code prediction model.
- Achieved a validation loss of **1.19** and a perplexity of **3.28** on held-out validation data.

Real-Time Emotion-Aware Multimodal Voice Assistant with Full-Duplex

Nov. 2025 – March 2026

- Technologies:** Python, FastAPI, WebSockets, OpenAI GPT-4o, Whisper, Deepgram, ElevenLabs TTS API.
- Architected a full-duplex conversational AI system using FastAPI and WebSockets for low-latency, real-time bi-directional audio and video streaming.
- Integrated a vision pipeline (MediaPipe/DeepFace) to analyze user emotion and demographics.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, R

Frameworks: React.js, CrewAI, Node.js, Flask, FastAPI, LangChain, PyTorch

AI & ML: LLMs, NLP, CUDA, Deep Learning, Prompt Engineering, Flash Attention, LLM's

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Kaggle, GitHub Actions, Hugging Face, Docker, UV

Libraries: Pandas, NumPy, Matplotlib, Uvicorn, PyTorch, TensorFlow, Scikit-learn

CERTIFICATIONS

Oracle Cloud Infrastructure (OCI) AI Foundations Associate — Oracle University (2025)

Data Analytics Virtual Experience Program — Deloitte (Forage)

HACKATHONS

Hangover Hackathon | WeMakeDevs, Cognee

July 2026

- Built **Oneiros**, a memory OS based on Erik Hoel's *Overfitted Brain Hypothesis*; creates an AI agent that sleeps to consolidate raw experiences and wakes up smarter with optimized, 3D-visualized long-term memory schemas.

Gemma 4 Good Hackathon | Google DeepMind

April 2026 – May 2026

- Built **LexiRead**, an AI-powered dyslexia-friendly reading assistant using Google's Gemma 4 family models.